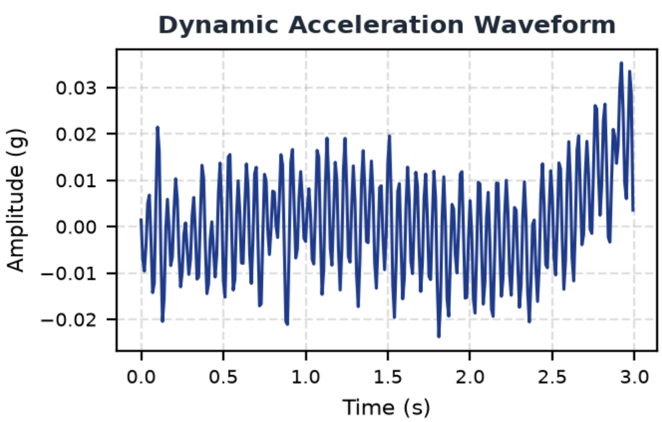
NEUROTRACK — Tremor Assessment & Telemetry Report

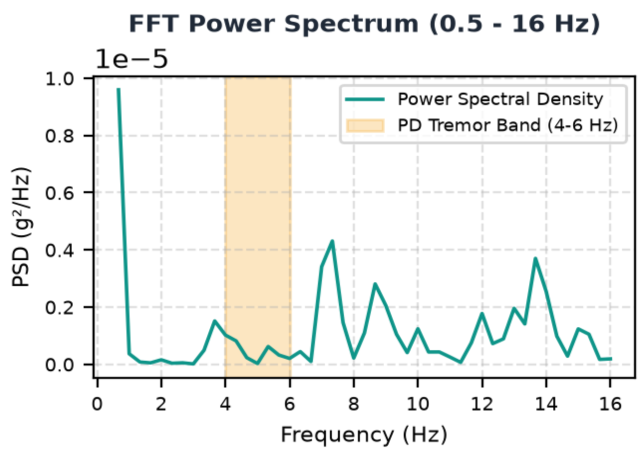
Report ID: NT-1788554308
Date: 2026-09-05 02:08:28

Patient Identifier:	LIVE_COM4	**Data Origin:**	Clinical Raw Recording
Sensor Device:	MPU6050 6-DoF IMU Ring	**Sampling Rate:**	100.0 Hz (Nyquist Band: 50 Hz)
Assessment Duration:	6.0 seconds	**Analysis Window:**	3.0 sec (50% sliding overlap)

AI Pattern Classification	Tremor Severity Index	Clinical Severity Grade
HEALTHY Confidence: 99.0% (PD Match: 0.0%)	**0.0** / 100	**Minimal / Negligible** Physiological baseline or voluntary movement. No rhythmic 4-6 Hz tremor detected.

Biomechanical Time-Series & Spectral Isolation





Extracted Digital Biomarkers (Window Metrics)

Biomarker	Observed Value	Reference Band / Diagnostic Context
Dominant Frequency	**0.00 Hz**	Parkinsonian Resting Tremor: 4.0 - 6.0 Hz
Tremor Band Power Ratio	**6.3%**	Ratio of power in 4-6 Hz band to broad spectrum (>40% indicates localized tremor)
Dynamic Jerk RMS	**1.03 g/s**	Time-derivative of acceleration measuring abrupt oscillatory changes
Spectral Shannon Entropy	**0.830**	Low (<0.45) indicates rhythmic periodicity; High (>0.70) indicates random/voluntary noise

Physiological Interpretation & Explainability

Dominant spectral energy is at 0.0 Hz, reflecting low-frequency voluntary movement or ambient baseline shifts rather than localized tremor. Tremor-band energy is minimal (6.3%), consistent with physiological resting state. Spectral entropy (0.83) reflects broad-spectrum kinetic noise or non-rhythmic movement.